

Lecture #2: Python for atomistic materials modeling

Previously on

- What is materials informatics
- Python crash course
- HW1 announcement

Goals

- Introduce python libraries for materials modeling
- Explain the advantages of using programming over the manual approach

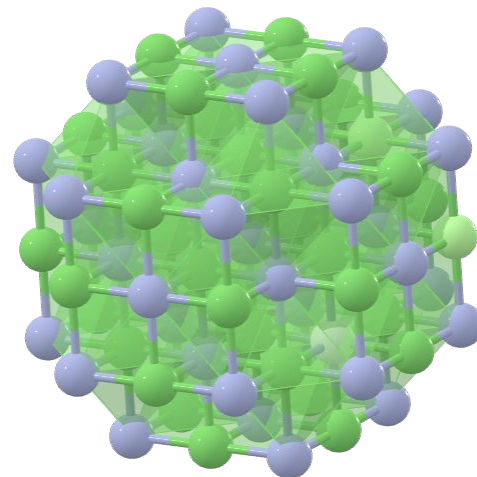
Agenda

- Materials modeling toy example
- Why use python?
- ASE and Pymatgen

Toy example: Band gap vs. X in cubic LiX, X = F, Cl, Br, I

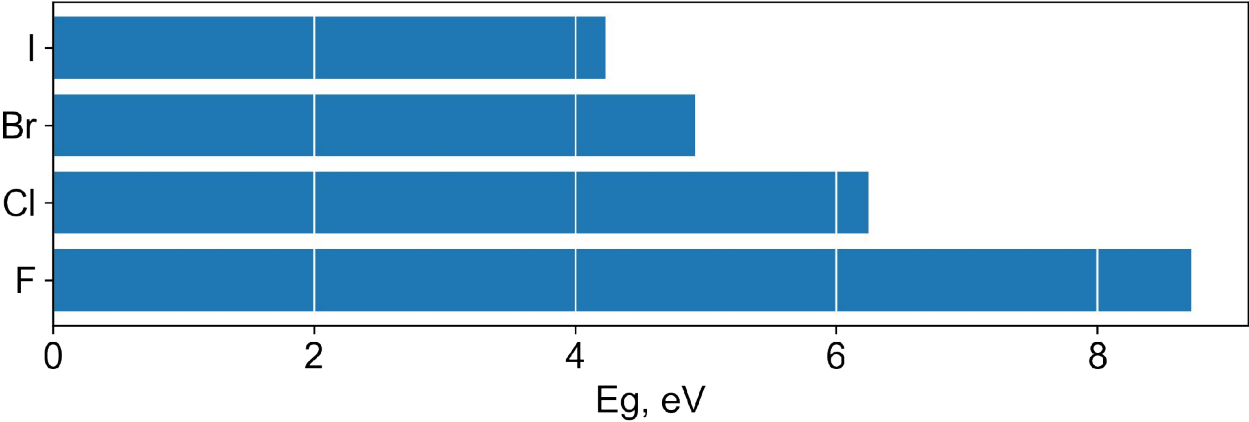
Task

- Calculate a band gap of LiX using a software “Y”
- Perform correlation analysis
- Explain the trend



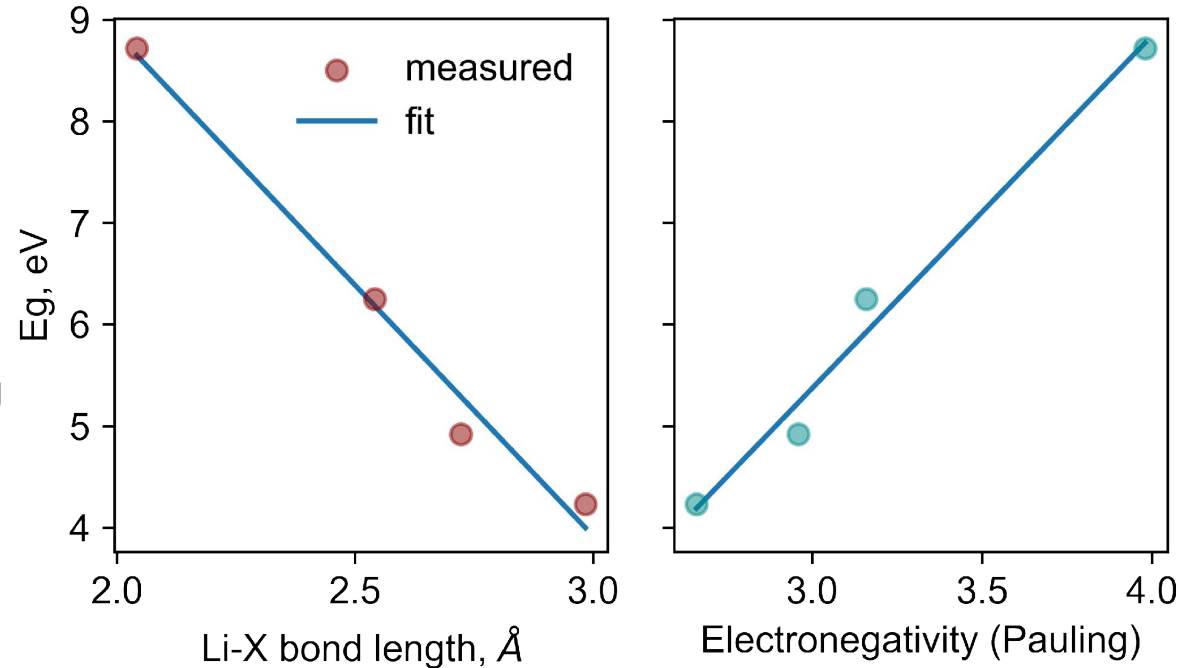
The trend

The higher atomic
number the lower E_g



Correlation analysis

- Li-X bond length vs. E_g
- Electronegativity (X) vs. E_g



Workflow

- Get(Create) structure file
- Prepare input files
- Run calculations
- Parse output
- Repeat the above steps for other structures
- Collect observables
- Analyze correlations
- Make conclusions

The software "Y" takes the following input files:

- structure file
- parameters of the calculation
- other specific files

... and outputs:

- 100,000 lines of text
- and one of these contains the E_g value

Which step takes the longest?

Which can be automated?

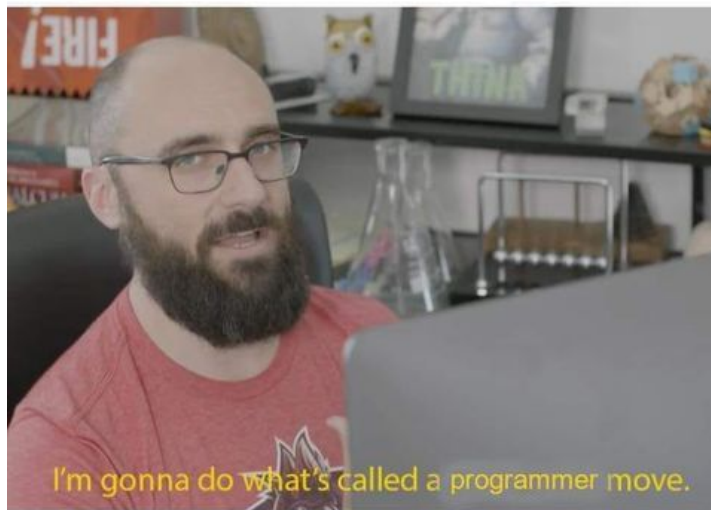
Manual vs. Automation

Manual

- Typing input files by hands
- Getting output data by opening files in text editors and copy/paste numbers
- Analysis of output data using Excel and Origin-like programs
- Relying on programs with graphical interface (in some cases its is efficient but in many cases you need to press many buttons every time to accomplish routine tasks)

Slow, easy to make a mistake, boring for repeating tasks

When there's a task that can be done manually in 10 minutes but you find a way to automate it in 10 days



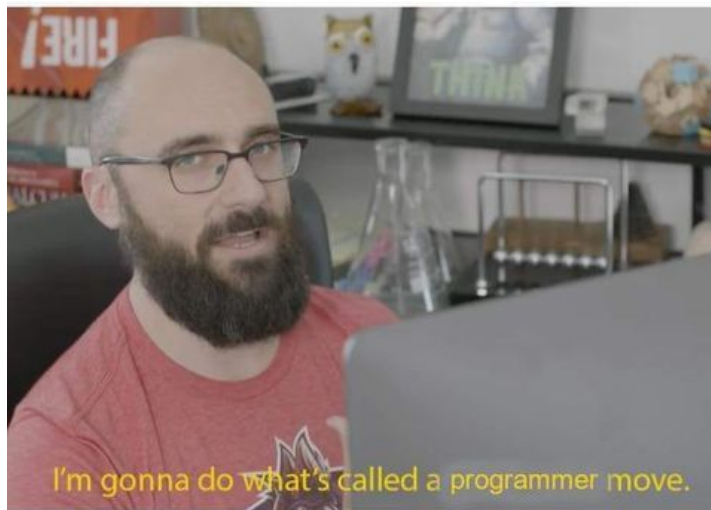
Manual vs. Automation

Automation

- All input files are created by scripts (only small changes are needed)
- Comprehensive and very flexible analysis specific for your needs
- Easy reuse of code for routine tasks
- Information can be stored in a database
- Takes more time in the beginning, but then is much faster

Fast, error-proof, fun

When there's a task that can be done manually in 10 minutes but you find a way to automate it in 10 days



Automation makes complex systematic studies possible

If you cannot program, you are fundamentally limited to small, manual studies. Programming is what turns computational materials science from “a few case studies” into a systematic, high-throughput science.

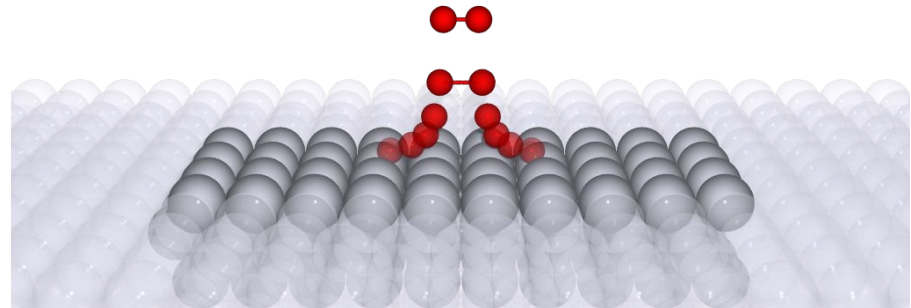
AI agents?

- Use AI agents to create automated workflows based on established tools, such as ASE and Pymatgen
- Use these workflows for production runs only after careful testing
- Do not use AI agents for doing stuff directly without established workflow - this leads to mistakes and errors due to hallucinations of AI agents

Python libraries for atomistic materials modeling:

ASE

- [ASE](#) - is an Atomic Simulation Environment written in the Python programming language with the aim of setting up, steering, and analyzing atomistic simulations.
- Good documentation with tutorials - Interfaces for most popular modeling simulation codes for density functional theory, molecular dynamics, etc
- Main features:
 - Atoms object for handling molecules/crystals
 - Read/write input files | Parse output files
 - Optimizers
 - Calculators
 - Visualization
 - Surface modeling

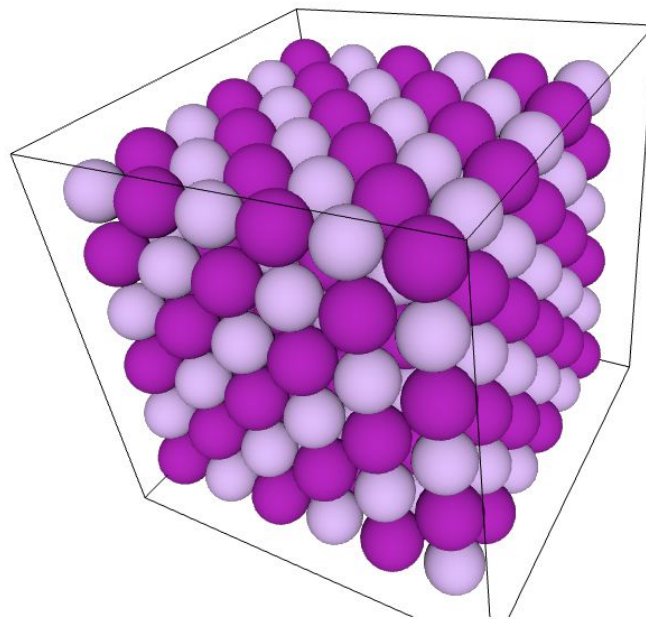


ASE minimum usage example

- read .cif file
- create supercell
- visualize
- save supercell
- more at the seminar

```
from ase.io import read, write
from ase.visualize import view
from ase.build import make_supercell

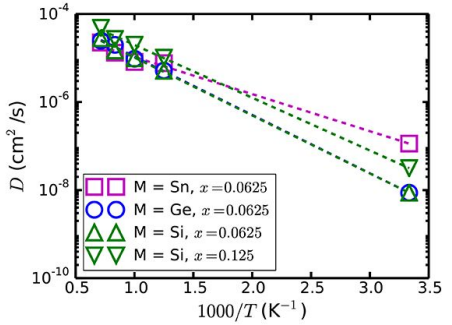
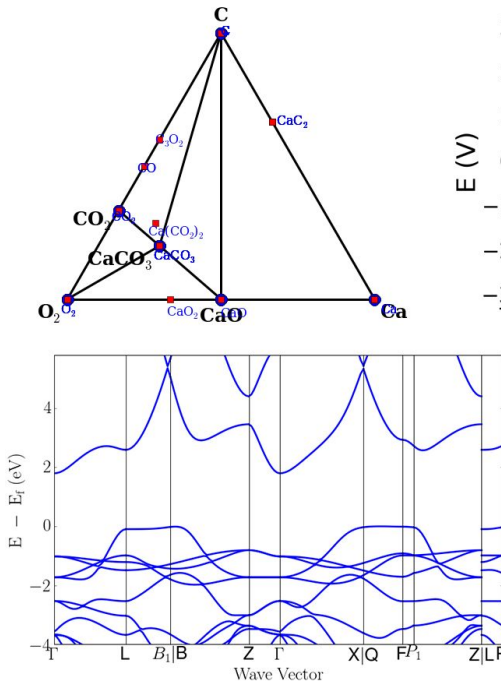
atoms = read('data/LiI.cif')
sc = make_supercell(atoms,
    [[3, 0, 0],
     [0, 3, 0],
     [0, 0, 3]]
)
view(sc, viewer='x3d')
write('data/LiI_3x3x3.cif', sc)
```



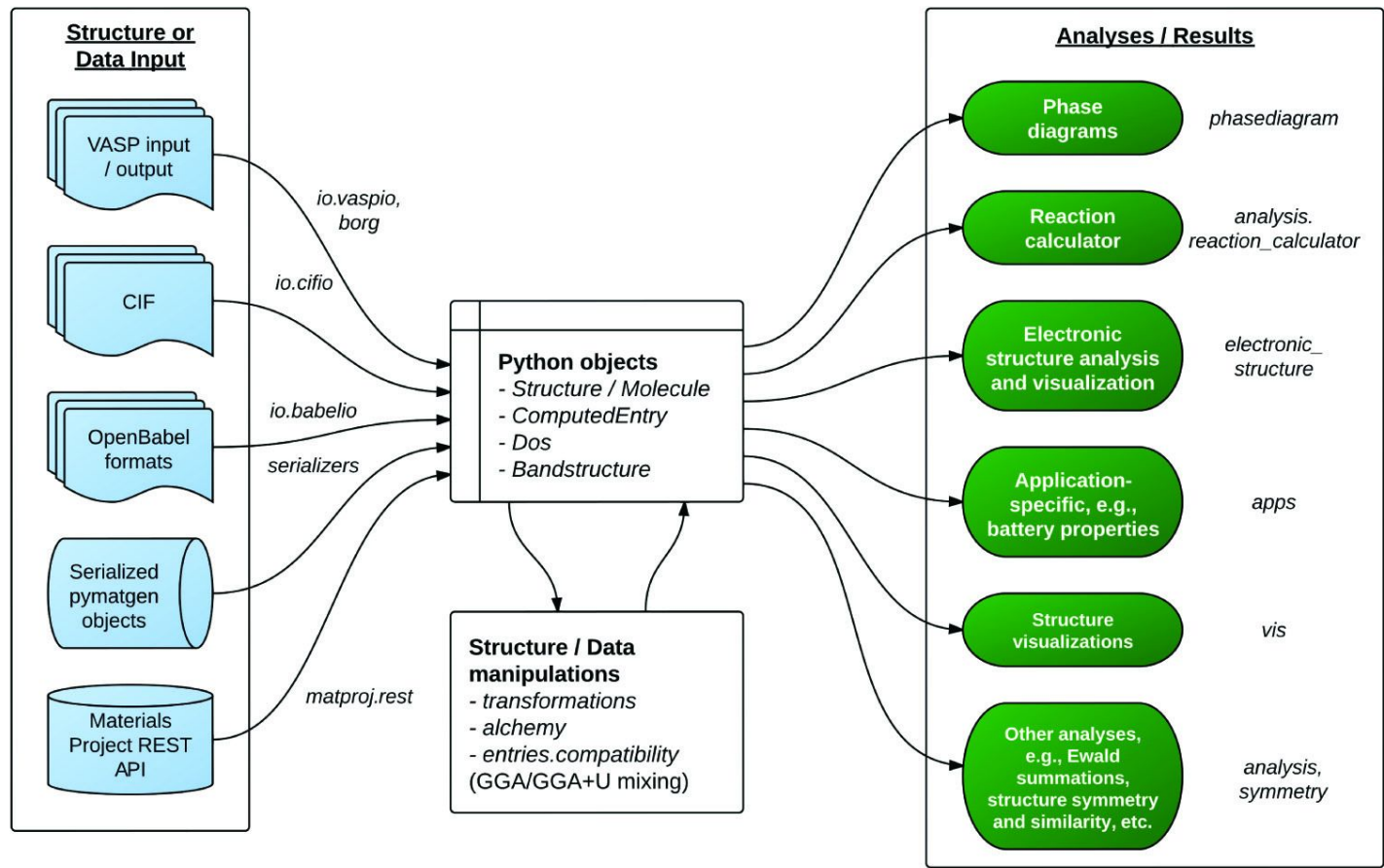
Python libraries for atomistic materials

modeling: Pymatgen

- [Pymatgen \(Python Materials Genomics\)](#) - a robust, open-source Python library for materials analysis
- Main features
- Highly flexible classes for the representation of Element, Site, Molecule, Structure objects.
- Powerful analysis tools, including generation of phase diagrams, Pourbaix diagrams, diffusion analysis, reactions, etc.
- Electronic structure analysis
- Integration with The Materials Project REST API (next lecture)



Basic workflow
with pymatgen



Take home message

Automation:

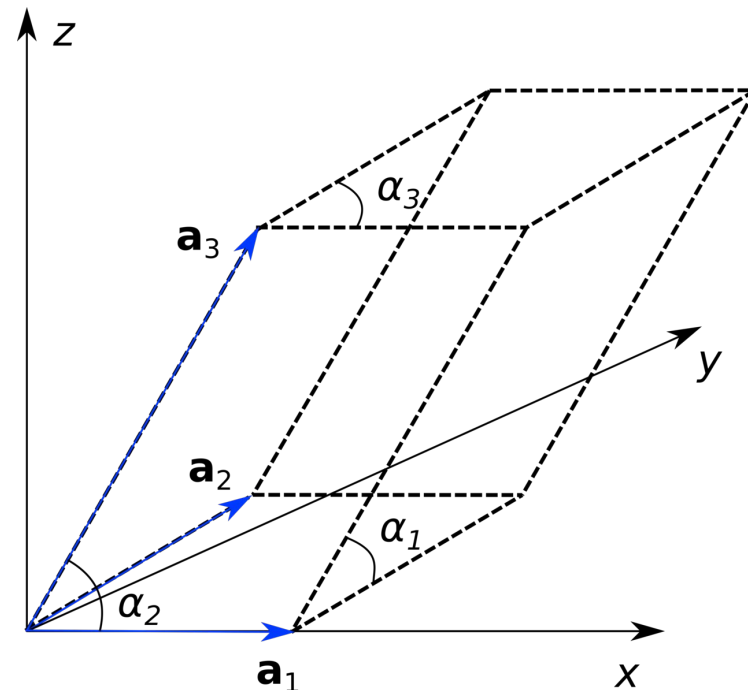
- saves your time
- gives you the opportunity (tools) to realize your ideas
- reduces number of mistakes (typos) made

A few words about periodic boundary conditions before the start of the seminar

Fractional coordinates:

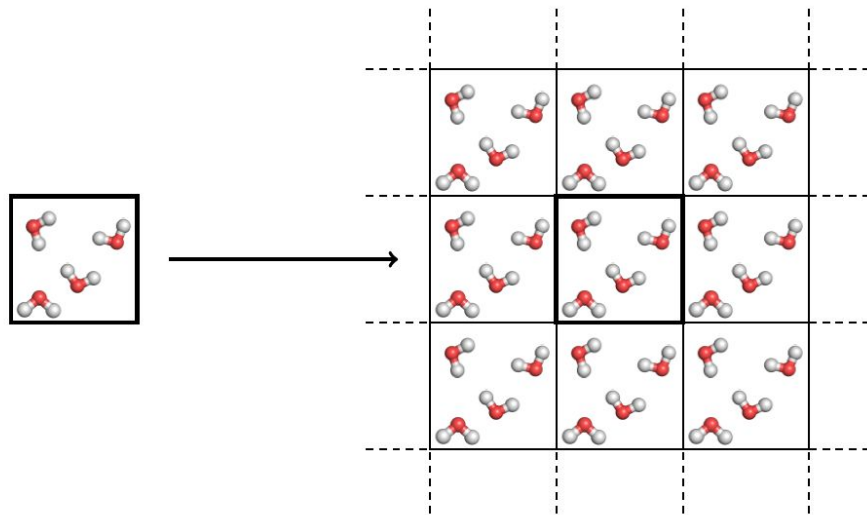
- Crystallographic basis ($\mathbf{a}_1$, $\mathbf{a}_2$, $\mathbf{a}_3$) = basis vectors of the coordinate system
- Coordinate of an atom: $\mathbf{r} = (x, y, z)$
- Fractional coordinate of the atom: $(x/|\mathbf{a}_1|, y/|\mathbf{a}_2|, z/|\mathbf{a}_3|)$

https://en.wikipedia.org/wiki/Fractional_coordinates



There is a three dimensional periodicity in crystals = translational symmetry

- The smallest repeating unit, **unit cell**, describes the whole crystal
 - translation (basis) vectors (**a**, **b**, **c**)
 - coordinates of atoms in the unit cell $\{\mathbf{r}_i(x, y, z)\}$
- Any atomic position, **r**, in the crystal is given by $\mathbf{r}_i^{mnk} = \mathbf{r}_i + m\mathbf{a} + n\mathbf{b} + k\mathbf{c}$
 - m, n, k - integer numbers



Thank you for your attention!